Science practice booklet: Grade 7 expectations

Grade 7 · Practice booklet

Name: _____

What this booklet is

This is general practice written for Science at Grade 7 level, following the Next Generation Science Standards standards for that grade. It is not made from your child's MAP questions, and it does not replace the classroom. Use it as a guide to what a child is expected to know and do at this level.

How to use it

- Do one section at a time, about 10 minutes, 3 days a week. Stop while it is still enjoyable.
- Read each question all the way through, twice, before answering. Underline the key words and say what is being asked.
- Take your time. A careful, correct answer is worth more than a fast one.
- Use scrap paper and show the working. Then check the answer before moving on.
- Ask your child to explain how they got an answer, even when it is right.
- If a section feels too hard, do the section before it again, or ask the teacher. That is part of learning, not a problem.

Draft: to be checked by a teacher before use.

Section 1: Living things

Standards practised: MS-LS3-1, MS-LS3-2, MS-LS4-4, MS-LS4-5

A worked example

A group of desert beetles has two colours: pale sand-coloured beetles and dark beetles. Birds find the dark beetles easily against the pale sand and eat more of them. Some beetles in each colour survive and have young. What is likely to happen to the colours of the beetles over many generations, and why?

1. Colour is an inherited trait, so young beetles tend to have the colour of their parents.
2. The pale beetles are hidden better, so more of them survive long enough to reproduce.
3. They pass on the genes for pale colour more often than the dark beetles pass on theirs.
4. Generation after generation, a larger share of the beetles will be pale. This is natural selection.

More and more of the beetles will be pale, because pale beetles survive and reproduce more often and pass on their genes.

Questions

1. What is a gene, and where are genes found?

2. Which of these are inherited traits and which are learned: eye colour, riding a bicycle, the shape of a seed, speaking Arabic?

3. In sexual reproduction, how many parents contribute genes to the offspring, and why are the offspring not exact copies of either parent?

4. What is asexual reproduction? Give one example from plants.

5. Date palms are either male trees or female trees. Pollen from a male flower must reach a female flower before dates can form. What is this process called, and which type of reproduction is it?

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6. Salem and his sister have the same parents, but they do not look exactly alike. Explain why.

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7. What is a mutation? Can a mutation be harmful, helpful or have no effect?

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8. The Arabian oryx has a pale coat, and a camel can go a long time without drinking. Why are these called adaptations?

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9. In a population of hares, some have longer legs and can run faster from a fox. Which hares are more likely to have young, and what may happen to leg length over many generations?

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10. Fossils show that many living things that lived long ago look different from living things today. What does this suggest about living things over a very long time?

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11. A farmer plants seeds only from the tomato plants that gave the biggest fruit, year after year. What is this called, and what is likely to happen to the size of the fruit?

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12. A long dry spell makes a plant food in an area very scarce. Explain, using the ideas of variation and natural selection, why a few individuals of a species may survive when many others do not.

Section 2: Earth and space

Standards practised: MS-ESS2-5, MS-ESS2-4, MS-ESS2-6, MS-ESS3-3, MS-ESS3-4, MS-ESS3-5

A worked example

The table gives the average yearly temperature in a city (example data): 1995: 26.8 C, 2005: 27.3 C, 2015: 27.8 C, 2025: 28.4 C. How much did the average temperature rise from 1995 to 2025, and what does this pattern suggest?

1. Subtract the first value from the last: 28.4 - 26.8 = 1.6.
2. Check the pattern: each reading is higher than the one before it.
3. A steady rise over thirty years is a trend, not a single hot year.
4. Scientists look for this kind of long-term trend, across many places, as evidence of a changing climate.

It rose by 1.6 C, a steady upward trend over 30 years.

Questions

1. What is the difference between a renewable and a non-renewable resource? Give one example of each.

2. Why is a solar park a good choice for electricity in the UAE?

3. Burning fuels such as petrol releases carbon dioxide. Why is this gas important for the climate?

4. Name two pieces of evidence that Earth's climate has been warming in recent decades.

5. What is the difference between weather and climate?

6. Warm air rises and cooler air sinks. Use this to explain why a sea breeze blows from the sea to the land on a sunny afternoon.

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7. What gives the energy that makes water evaporate from the sea in the water cycle?

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8. Put these steps of the water cycle in order: precipitation, evaporation, condensation.

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9. Desalination plants give the UAE fresh water, but they use a lot of energy. Give one way to reduce how much water we need to make.

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10. Mangrove trees take in carbon dioxide and protect the coast. Give two reasons why planting more mangroves is a good way to care for the environment.

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11. As the number of people in a city grows, what usually happens to the amount of water, food and energy the city needs? Why?

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12. Compare two ways of getting to work: by car alone and by metro. Which usually puts less carbon dioxide into the air for each person, and why?

Section 3: Matter, forces and energy

Standards practised: MS-PS1-1, MS-PS1-2, MS-PS1-5, MS-PS2-1, MS-PS2-2

A worked example

Hamdan seals a fizzy tablet and some water inside a strong bag. The bag and its contents have a mass of 150 g. The tablet fizzes and the bag swells with gas. What is the mass of the sealed bag after the reaction, and why?

1. A chemical reaction rearranges atoms into new substances.
2. No atoms are made or destroyed, and nothing can escape from a sealed bag.
3. So the total mass stays the same: it is still 150 g.
4. The gas is still inside the bag. If the bag were open, the gas would escape and the scale would show less.

150 g, because mass is conserved: the atoms are only rearranged and none left the bag.

Questions

1. What is the difference between an atom and a molecule?

2. A water molecule is written H2O. How many atoms of hydrogen and how many atoms of oxygen does one molecule have?

3. Name two signs that a chemical reaction may have taken place.

4. Iron nails left outside become covered in orange-brown rust. Is rust the same substance as iron? How do you know?

5. 4 g of hydrogen react completely with 32 g of oxygen to make water. What is the mass of the water made?

6. A reaction begins with 8 atoms. How many atoms are there after the reaction? Explain.

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7. Noura rides her bicycle 36 km in 3 hours at a steady speed. What is her speed?

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8. A book rests on a table. Gravity pulls it down, yet it does not move. Explain using forces.

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9. A ball rolls across a classroom floor and gradually stops. Which force makes it stop, and are the forces on it balanced?

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10. A bus brakes suddenly and the passengers lean forward. Use Newton's first law to explain.

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11. Khalid pushes an empty shopping trolley and then pushes the same trolley full of shopping, using the same force. Which one speeds up more, and why?

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12. A rocket pushes hot gas downwards and the rocket moves upwards. Which of Newton's laws does this show, and what does it say?

Answer guide for parents

Section 1: Living things

1. A gene is a section of DNA that carries instructions for a trait. Genes are found on chromosomes inside the nucleus of a cell.
2. Inherited: eye colour and the shape of a seed. Learned: riding a bicycle and speaking Arabic.
3. Two parents each give half of the genes, so the offspring get a new mixture of genes and are different from both parents.
4. Reproduction with only one parent, so the offspring have the same genes as the parent. Example: a new date palm that grows from an offshoot of a parent palm.
5. Pollination. It is sexual reproduction, because genes from two parents are combined in the seed.
6. Each child gets a different mixture of the genes from the two parents, so there is variation between brothers and sisters.
7. A mutation is a change in a gene. It can be harmful, helpful or have no effect on the living thing, depending on the change and the environment.
8. An adaptation is an inherited feature that helps a living thing survive in its environment. The pale coat helps reflect sunlight and the camel's body uses water carefully, which suits the hot, dry desert.
9. The faster hares survive longer and have more young. Over many generations more hares in the population will have long legs.
10. Living things change over very long periods of time, and many kinds that once lived are now extinct.
11. It is selective breeding (artificial selection). The fruit is likely to become bigger over time, because people choose which plants pass on their genes.
12. Individuals of a species vary. Some may have traits, such as needing less food or water, that suit the new conditions, so they survive and pass on those genes to their young.

Section 2: Earth and space

1. A renewable resource is replaced naturally in a short time (for example sunlight or wind). A non-renewable resource takes millions of years to form, so it can run out (for example oil or natural gas).
2. The UAE has a lot of strong sunshine, sunlight is renewable, and solar panels make electricity without burning fuel.
3. Carbon dioxide is a greenhouse gas. It traps heat in the atmosphere, so adding more of it tends to make Earth warmer.
4. Any two of: rising global average temperatures, melting glaciers and ice sheets, rising sea levels, warmer oceans.
5. Weather is the condition of the atmosphere in one place at one time (such as hot and sunny today). Climate is the usual pattern of weather in a place over many years.
6. The land heats up faster than the sea, so the air over the land warms and rises. Cooler air from over the sea moves in to replace it, and this moving air is the sea breeze.

7. The energy from the Sun.
8. Evaporation, then condensation, then precipitation (water vapour rises, cools into clouds, and falls as rain).
9. For example: fix leaks, use less water at home, reuse treated water for irrigation. (Any sensible answer.)
10. They take carbon dioxide from the air, shelter fish and birds, and protect the shore from waves. (Any two.)
11. The need grows, because more people use more resources every day.
12. The metro, because many people share one train, so less fuel or electricity is used for each person.

Section 3: Matter, forces and energy

1. An atom is a very small particle that makes up matter. A molecule is two or more atoms joined together.
2. 2 hydrogen atoms and 1 oxygen atom (3 atoms in all).
3. Any two of: a gas is given off (bubbles), a colour change, a new solid forms, heat or light is given out.
4. No. Rusting is a chemical reaction in which iron reacts with oxygen (and water) to make a new substance with different properties, such as colour and hardness.
5. 36 g (4 + 32; mass is conserved in a chemical reaction)
6. 8 atoms. The atoms are rearranged into new substances, but none are made or destroyed.
7. 12 km per hour (36 divided by 3)
8. The table pushes up on the book with a force equal to gravity pulling down. The forces are balanced, so there is no change in motion.
9. Friction between the ball and the floor. The forces are unbalanced, so the ball's motion changes.
10. An object keeps moving at the same speed in a straight line unless an unbalanced force acts on it. The bus slows, but the passengers' bodies tend to keep moving forward.
11. The empty trolley. For the same force, an object with less mass speeds up more.
12. Newton's third law: when one object pushes on another, the second pushes back on the first with an equal force in the opposite direction.